



Iterative Complement-clustering PCA: Uncovering latent industry structures in stock returns

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ABSTRACT

Principal component analysis (PCA) is a widely implemented statistical method for dimension reduction, but struggles to identify group-specific patterns (sub-homogeneity), such as the latent industry structures in high-dimensional stock return data. We propose an Iterative Complement-clustering PCA (ICcPCA) that jointly estimates homogeneity (market-wide effects) and sub-homogeneity (industry-specific risks), where a Leave-one-out principal component regression (LOO-PCR) clustering approach is developed to iteratively cluster variables (stocks) into disjoint multidimensional subspaces (groups). Simulations show that the ICcPCA outperforms the conventional PCA in both estimating the number of principal components and recovering the data. In analyzing stock returns of 160 firms across 8 industries, ICcPCA with LOO-PCR can separate market-wide effects from industry-specific risks, achieving higher clustering accuracy and lower recovering errors. Applications in portfolio optimization demonstrate that ICcPCA-based minimum variance portfolios can attain lower volatility and higher profitability than PCA-based portfolios.

1. Introduction

PCA (Jolliffe, 2002; Anderson, 2003) is a dominant tool for dimension reduction in economic and financial studies, yet its inability to uncover latent group-specific patterns in heterogeneous data remains a critical limitation. For instance, stock returns exhibit both market-wide homogeneity (e.g., macroeconomic shocks) and industry-specific sub-homogeneity (e.g., sectoral regulations or supply chain disruptions, Fama and French, 1997). While methods like principal orthogonal complement thresholding (POET, Fan et al., 2013) address cross-sectional dependence, they overlook the intrinsic cluster structure of variables. Advances in structured factor models (Bai and Ng, 2019) and high-dimensional clustering (Bonhomme and Manresa, 2015) highlight the need for methods that simultaneously estimate common factors and cluster-specific components, which is the primary goal of this paper.

In this work, we propose an Iterative Complement-clustering PCA (ICcPCA) to jointly estimate market-wide homogeneity and industry-specific sub-homogeneity. Our method iterates between two steps: (1) extracting common factors via a hierarchical PCA that aggregates cluster-specific components, and (2) refining clusters using a novel Leave-one-out principal component regression (LOO-PCR) approach. This contrasts with thresholding-based methods like POET (Fan et al., 2013) and enhances structured factor models (Bai and Ng, 2019) by

integrating factor analysis with data-driven group identification. The advantage of incorporating a clustering step is twofold, and the first one is on its economic and financial implications. Our approach can identify groups of companies whose stock returns are highly correlated, which can help researchers and investors understand the stock market. Meanwhile, as a statistical method, the clustering involved in ICcPCA can also improve the estimation accuracy of the covariance matrix, which promotes applications such as creating minimum variance portfolios (MVP). Simulation studies and analysis on stock returns in the US market confirm that ICcPCA outperforms PCA and POET on separate industry-specific sub-homogeneity from market-wide homogeneity. Additionally, the ICcPCA-based MVP also attains lower volatility and higher profitability than PCA-based portfolios. Due to space limitations, some technical details and additional statistical applications of ICcPCA are left to the online Supplementary Material.

2. Method

Let $i = 1, \dots, n$ be the index of the sample and $l = 1, \dots, p$ be the index of the variable. Consider for the data $\mathbf{x}_i = [x_{1i}, \dots, x_{pi}]^T \in \mathbb{R}^p$, conventional PCA assumes a low-rank structure as

$$\mathbf{x}_i = \Phi \mathbf{g}_i + \mathbf{u}_i, \quad (1)$$

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where $\mathbf{g}_i = [g_{i1}, \dots, g_{ir_c}]^\top$ is an $r_c \times 1$ vector contains the first r_c principal components, $\Phi = [\phi_1, \dots, \phi_p]^\top$ is a $p \times r_c$ matrix of eigenvectors for the first r_c principal components and $\mathbf{u}_i$ is the noise. When sub-homogeneity exists, the low rank model (1) becomes

$$\mathbf{x}_i = \Phi \mathbf{g}_i + \sum_{j=1}^J \Gamma^{(j)} \mathbf{f}_i^{(j)} + \sum_{j=1}^J \Gamma^{*(j)} \mathbf{e}_i^{(j)}, \quad i = 1, \dots, n, \quad j = 1, \dots, J, \quad (2)$$

where p variables are split into J clusters each with p_j variables such that $p = \sum_{j=1}^J p_j$, $\mathbf{f}_i^{(j)} = [f_{i1}^{(j)}, \dots, f_{ir_j}^{(j)}]^\top$ is an $r_j \times 1$ vector containing the j th cluster-specific components capturing the sub-homogeneity of j th group with $\Gamma_{p \times r_j}^{*(j)} = [\mathbf{0}_{r_j \times p_1}^{(j)}, \dots, \mathbf{I}_{r_j \times p_j}^{(j)}, \dots, \mathbf{0}_{r_j \times p_J}^{(j)}]^\top$ containing eigenvectors $\Gamma_{p_j \times r_j}^{(j)} = [\gamma_1^{(j)}, \dots, \gamma_{p_j}^{(j)}]^\top$ corresponding to $\mathbf{f}_i^{(j)}$, and $\mathbf{e}_i^{(j)} = [e_{i1}^{(j)}, \dots, e_{ip_j}^{(j)}]^\top$ is an $p_j \times 1$ vector of noise in the j th group with $\Gamma_{p \times p_j}^{*(j)} = [\mathbf{0}_{p_j \times p_1}^{(j)}, \dots, \mathbf{I}_{p_j \times p_j}^{(j)}, \dots, \mathbf{0}_{p_j \times p_J}^{(j)}]^\top$ a block matrix containing the identity matrix $\mathbf{I}_{p_j}^{(j)}$.

Model (2) can also be written in the matrix form for $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n]^\top \in \mathbb{R}^{n \times p}$ as

$$\mathbf{X} = \mathbf{G} \Phi^\top + \sum_{j=1}^J \mathbf{F}^{(j)} \Gamma^{*(j)\top} + \sum_{j=1}^J \mathbf{e}^{(j)} \Gamma^{*(j)\top}, \quad j = 1, \dots, J, \quad (3)$$

where $\mathbf{G} = [\mathbf{g}_1, \dots, \mathbf{g}_n]^\top$ represents the homogeneity, $\mathbf{F}^{(j)} = [\mathbf{f}_1^{(j)}, \dots, \mathbf{f}_n^{(j)}]^\top$ is the sub-homogeneity within the j th cluster and $\mathbf{e}^{(j)} = [\mathbf{e}_1^{(j)}, \dots, \mathbf{e}_n^{(j)}]^\top$. For the identification of both the homogeneity and sub-homogeneity in model (2), we introduce the following conditions.

Assumption 1. (1) The homogeneity satisfy $\mathbb{E} \|\mathbf{g}_i\|^4 < \infty$ and $n^{-1} \sum_{i=1}^n \mathbf{g}_i \mathbf{g}_i^\top \rightarrow \Sigma_G$ as $n \rightarrow \infty$, where Σ_G is an $r_c \times r_c$ positive definite matrix. The group-specific sub-homogeneity satisfies $\mathbb{E} \|\mathbf{f}_i^{(j)}\|^4 < \infty$ and $n^{-1} \sum_{i=1}^n \mathbf{f}_i^{(j)} \mathbf{f}_i^{(j)\top} \rightarrow \Sigma_{F^{(j)}}$ as $n \rightarrow \infty$, where $\Sigma_{F^{(j)}}$ is an $r_j \times r_j$ positive definite matrix for $j = 1, \dots, J$. In addition, $\mathbf{f}_i^{(j_1)}$ and $\mathbf{f}_i^{(j_2)}$ ($j_1 \neq j_2$) are not perfectly correlated, and $n^{-1} \sum_{i=1}^n \mathbf{g}_i \mathbf{f}_i^{(j)\top} = \mathbf{0}$ for $j = 1, \dots, J$.

(2) The eigenvectors fulfill $\mathbb{E} \|\phi_i\|^4 < \infty$ and $\|\frac{1}{p} \Phi^\top \Phi - \Sigma_\Phi\| \rightarrow 0$ as $p \rightarrow \infty$, where Σ_Φ is an $r_c \times r_c$ positive definite matrix. And $\mathbb{E} \|\Gamma_i^{(j)}\|^4 < \infty$, $\|\frac{1}{p_j} \Gamma_i^{(j)\top} \Gamma_i^{(j)} - \Sigma_{\Gamma^{(j)}}\| \rightarrow 0$ as $p_j \rightarrow \infty$, where $\Sigma_{\Gamma^{(j)}}$ is an $r_j \times r_j$ positive definite matrix, $j = 1, \dots, J$. In addition, we assume $\frac{p^{1-\delta}}{n} \rightarrow 0$ and $\frac{p_j^{1-\zeta_j}}{n} \rightarrow 0$, as $p, p_j, n \rightarrow \infty$. Here $0 < \zeta_j < \delta \leq 1$.

(3) for each j and l , $\mathbf{g}_i^\top \phi_l$ and $\mathbf{f}_i^{(j)\top} \gamma_l^{(j)}$ are strong mixing processes with mixing coefficients satisfying $r(t) \leq \exp(-a_1 t^{b_1})$ and with tail probabilities $P(\|\mathbf{g}_i^\top \phi_l\| > z) \leq \exp\{1 - (z/b_2)^{a_2}\}$ and $P(\|\mathbf{f}_i^{(j)\top} \gamma_l^{(j)}\| > z) \leq \exp\{1 - (z/b_3)^{a_3}\}$, where a_1, a_2, a_3, b_1, b_2 , and b_3 are positive constants.

(4) $\mathbb{E}[\epsilon_{il}^{(j)}] = 0$, $\text{var}(\epsilon_{il}^{(j)}) = \sigma^{(j)2}$, $\epsilon_{il}^{(j)}$ is independent over i and l . A positive constant $C < \infty$ exists such that $\mathbb{E}[\|\epsilon_{il}^{(j)}\|^8] < C$ for all i and l . $\epsilon_{il}^{(j)}$ is independent of $\mathbf{g}_s, \phi_l, \mathbf{f}_s^{(j)}$ and $\gamma_l^{(j)}$ ($j = 1, \dots, J$) for all i, l, t, s .

(5) The number of members in cluster j fulfills $0 < \underline{a} < p_j/p < \bar{a} < 1$. Here $\underline{a}$ and $\bar{a}$ are two constants.

The conditions in Assumption 1 imply the existence and identification of both the homogeneity and the sub-homogeneity. Similar conditions are also considered by, e.g., Ando and Bai (2017), Wang and Fan (2017), where the estimation consistency is studied.

In general, there are two goals in estimating model (3). In the presence of an unknown cluster structure, the first goal is to cluster the variables into disjoint multidimensional subspaces (i.e., determine which variables are in the same group). The second goal is to correctly estimate both the common components $\mathbf{G}$ and the cluster-specific components $\mathbf{F}^{(j)}$ with their corresponding eigenvectors. These components serve as a low-rank representation of the data and can therefore be used for further applications such as estimating the covariance matrix of stock returns to construct portfolios with minimum variance. Further discussions are left to Section 4.

Correctly estimating $\mathbf{G}$ and $\mathbf{F}^{(j)}$ poses several challenges. Due to the ‘Curse of dimensionality’, PCA often performs very poorly when

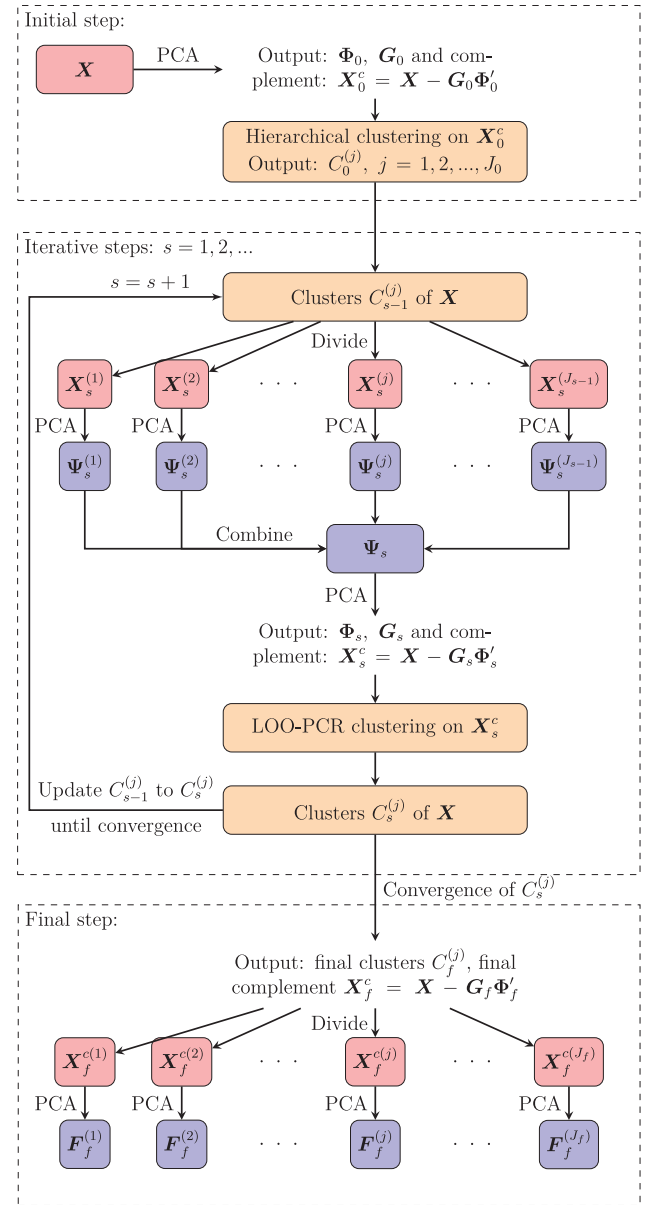


Fig. 1. Flowchart of ICcPCA.

p is much larger than n , as a large p may blur the spike structure of the sample covariance matrix. The group structure of variables then motivates us to separate the data into clusters and extract the common information from each cluster to determine the homogeneity. However, the dissimilarity between clusters of variables is often masked by the homogeneity. Hence to accurately identify the clusters, we must remove the homogeneity first and then cluster variables based on the complement ($\mathbf{X}^c = \mathbf{X} - \mathbf{G}\Phi^\top$), whereas $\mathbf{G}$ and Φ are not known in advance. This inspires us to propose a novel iterative method, ICcPCA, with an LOO-PCR clustering approach to cluster the variables and estimate the components simultaneously. A flowchart illustrating the method is presented in Fig. 1, and the detailed procedure is described in Algorithm 1 in Appendix A.

The key contribution of our algorithm is iteratively estimating the common components and cluster-specific components. Motivated by the group structure of the data, we utilize the Iterative Steps in ICcPCA to first cluster variables and then perform PCA in each cluster, followed by a second layer of PCA that extracts common information shared by all clusters. The cluster information is then updated and improved by a

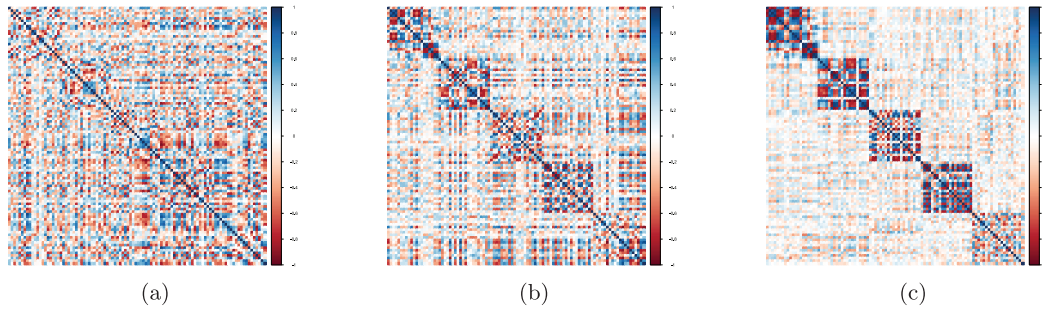


Fig. 2. Correlation plot using data generated by simulation Example 2 for (a) the original data X , (b) the initial complement $X_0^c = X - G_0 \Phi_0^T$ and (c) the final complement $X_f^c = X - G_f \Phi_f^T$.

novel LOO-PCR clustering approach inspired by [Chiou and Li \(2007\)](#). As ICcPCA is performed on the same data iteratively, to avoid overfitting, a ratio-based estimator ([Lam and Yao, 2012](#)) with an upper limit is used to select the number of principal components, and similarly, an upper threshold of the number of clusters is also considered when determining the membership in LOO-PCR clustering (see [Appendix A](#) for details).

ICcPCA with LOO-PCR clustering is more effective in estimating the common components as the PCA is performed with a smaller p . An example illustrating the performance of ICcPCA and LOO-PCR clustering is shown in [Fig. 2](#) with X generated by simulation Example 2. [Fig. 2\(a\)](#) presents the correlation plot for the original data X where the cluster structure is masked by the homogeneity. [Fig. 2\(b\)](#) demonstrates that after removing the homogeneity estimated in the Initial Step, it is still difficult to perceive the cluster structure. However, if we remove the homogeneity estimated in the Final Step of ICcPCA, the block-diagonal structure is clear, implying a prominent heterogeneity within the data. Therefore, iteratively estimating the common components is advantageous.

3. Simulation studies

In this section, we conduct various simulation studies to investigate the performance of our proposed ICcPCA compared with the conventional PCA under different simulation settings.

3.1. Simulation settings

The data-generating process is based on model (2). Specifically, we first generate orthonormal matrices Φ and $\Gamma^{(j)}$ as eigenvectors corresponding to common and cluster-specific components, respectively. Secondly, we generate $\{g_{ik}\}_{k=1}^{r_c}$ from $\sqrt{\delta_k} W_{ik}$, with W_{ik} i.i.d. $\mathcal{N}(0, 1)$ and eigenvalues $\{\delta_k\}_{k=1}^{r_c}$ defined as $\delta_1 > \dots > \delta_{r_c} > 0$. Similarly, for each cluster j , we generate $\{f_{im}^{(j)}\}_{m=1}^{r_j}$ from $\sqrt{\lambda_m^{(j)}} Z_{im}^{(j)}$, with $Z_{im}^{(j)}$ i.i.d. $\mathcal{N}(0, 1)$ and eigenvalues $\{\lambda_m^{(j)}\}_{m=1}^{r_j}$ defined as $\lambda_1^{(j)} > \dots > \lambda_{r_j}^{(j)} > 0$. Lastly, we generate errors of $\epsilon_i^{(j)}$ as i.i.d. $\mathcal{N}(0, \sigma^{(j)2})$.

In our simulation study, we consider four different examples, where training and test sets are generated with the same sample size.

- **EXAMPLE 1** ($n = 50$, $p = 100$). $r_c = 3$, $\delta_k \sim \mathcal{N}(125, 5)$, $J = 5$, $p_j = 20$, $r_j = 2$, $\lambda_m^{(j)} \sim \mathcal{N}(\theta_{j,h}, 1)$ with $\theta_{1,1} = \theta_{1,2} = 25$, $\theta_{5,1} = \theta_{5,2} = 5$, $\theta_{j,1} = 25$ and $\theta_{j,2} = 5$ for $j = 2, 3, 4$. $\sigma^{(j)} = 0.5$ for $j = 1, 2, \dots, 5$.
- **EXAMPLE 2** ($n = 30$, $p = 100$). This example is identical to Example 1, except that the sample size is decreased from 50 to 30.
- **EXAMPLE 3** ($n = 50$, $p = 200$). $r_c = 3$, $\delta_k \sim \mathcal{N}(125, 5)$, $J = 10$, $p_j = 20$, $r_j = 2$, $\lambda_m^{(j)} \sim \mathcal{N}(\theta_{j,h}, 1)$ with $\theta_{1,1} = \theta_{1,2} = \theta_{2,1} = \theta_{2,2} = 25$, $\theta_{9,1} = \theta_{9,2} = \theta_{10,1} = \theta_{10,2} = 5$, $\theta_{j,1} = 25$ and $\theta_{j,2} = 5$ for $j = 3, \dots, 8$. $\sigma^{(j)} = 0.5$ for $j = 1, \dots, 10$. (This example doubles the number of clusters in Example 1, with other settings remaining the same.)

- **EXAMPLE 4** ($n = 30$, $p = 100$). This example is identical to Example 2, but without the homogeneity, which is a special case of model 2.

For all examples, we repeat the simulation 100 times and investigate the performance of ICcPCA in terms of the identifiability of the number of PCs and recovering accuracy. The recovering accuracy is measured by the mean squared recovering error (MSRE) in the test set computed as $\frac{1}{np} \|\hat{X}_{test} - X_{test}\|_F^2$, where $\hat{X}_{test}$ is computed using PCs and eigenvectors extracted from the training set.

3.2. Simulation results

The boxplots of the total number of PCs selected (No. PCs) and MSRE for Examples 1 to 4 are presented in [Fig. 3](#). To demonstrate the effect of LOO-PCR, we denote the ICcPCA with initial clusters as ICcPCA_I and the ICcPCA with LOO-PCR clusters as ICcPCA_F, and compare the results of the two approaches. As shown in [Fig. 3](#), ICcPCA_I estimates a consistently higher number of PCs than its iterative counterpart (ICcPCA_F) and the true number of PCs (dotted line), which implies the advantages of using the iterative estimation procedure.

From the recovering accuracy perspective, it is worth noting from [Fig. 3](#) that PCA always performs poorly under this cluster structure of the data, while both ICcPCA-based methods demonstrate superior performance in recovering the data because they achieve lower MSRE than PCA. In general, ICcPCA_F results in more accurate partitions and produces more reliable estimations of the cluster-specific components, especially for those with small variations, which contribute the most to predicting the response according to our simulation settings. This further confirms the advantages of using our proposed method.

Lastly, in Example 4, ICcPCA_I and ICcPCA_F simply reduce to performing PCA on each cluster of variables based on hierarchical clustering and LOO-PCR clustering, respectively. From the last row of [Fig. 3](#), we observe that applying PCA after clustering the variables demonstrates better performance than applying PCA directly to the data. And ICcPCA_F achieves a better estimate of the number of PCs than ICcPCA_I, implying that our proposed LOO-PCR clustering outperforms hierarchical clustering.

4. Analysis on stock returns data

For further illustration, we study a dataset of stock returns using the proposed ICcPCA and compare it with the conventional PCA. In particular, we are interested in the performance of our proposed ICcPCA on clustering stock returns of companies from various industries.

4.1. Clustering stock returns

The data are collected from the Center for Research in Security Prices (CRSP) and include the daily stock returns of 160 companies from 1st January 2014 to 31st December 2014, with 252 trading days. The 160 stocks are selected from eight different industries according to

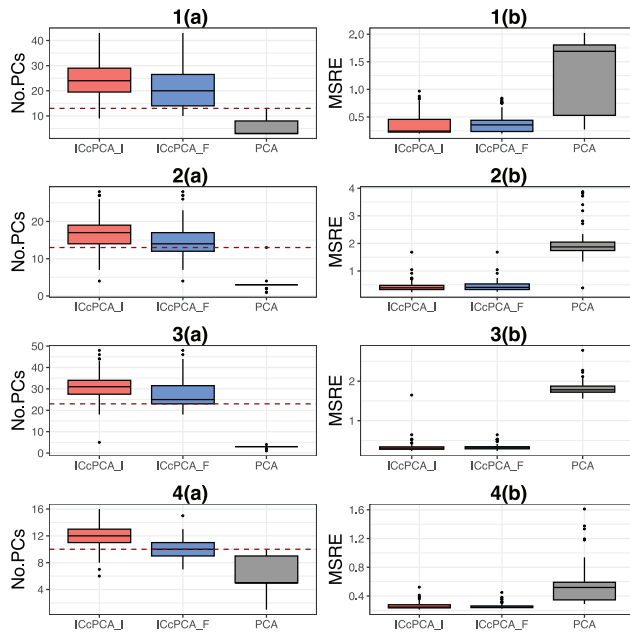


Fig. 3. Boxplots of (a) estimates of the number of PCs (No. PCs) with dotted lines indicating the true numbers, and (b) MSRE, based on 100 simulations of Examples 1 to 4.

Fama and French's 48-industry classification (Fama and French, 1997), with 20 stocks from each industry. The data for the first 126 trading days are treated as the training set, and the rest are the test set. In particular, we aim to identify the clusters that consist of companies from the same industry. This cluster structure of stock returns is also discussed in Fan et al. (2013).

Fig. 4(a) displays the correlation plot for the original data. A vague cluster structure can be observed, but the off-diagonals are clearly non-zero, implying that stocks from different industries tend to be positively correlated. We apply hierarchical clustering directly to the original data and find that most stocks from different industries are clustered in the same group, as displayed in Fig. 5(b). Only stocks from the 5th industry stand out as another cluster, because they are less correlated with other stocks.

Next, we apply ICcPCA to stock returns, and one common component is determined. Fig. 4(b) presents the correlation plot of the data after the homogeneity is removed. Comparatively, the clusters are more apparent. One can interpret the common components (i.e. the homogeneity) as the market return and the rest of the components (i.e. the sub-homogeneity) as industry-specific returns. As demonstrated in Fig. 5(c), after the Initial Step of ICcPCA, a few clusters of stocks from the 1st, 2nd, and 7th industries are correctly identified, while stocks from other industries (the 3rd, 4th, 5th, 6th, and 8th) cannot be correctly separated from two large clusters. Fig. 5(d) depicts the final clusters after the Iterative Step of ICcPCA, where most stocks from the aforementioned two large clusters can be correctly separated according to industries with a minor cost of a few miss-classifications in the 1st and 7th cluster. The final clusters mainly capture the industry information, but a few stocks are not clustered into their own industries. This is to be expected as some stocks from the same industry are not necessarily highly correlated, as shown in Fig. 4 (e.g., some stocks from the 1st, 3rd, 6th, and 7th industries), due to diversification of enterprises.

From the prediction perspective, we also compute the MSRE for PCA, ICcPCA_I, and ICcPCA_F on the test set as 3.56, 2.23, and 1.67 (in the unit of 10^{-4}), respectively. In other words, ICcPCA_F can not only capture the market effect of stock returns but also include some industrial risks, hence achieving a higher out-of-sample prediction accuracy on high-dimensional stock returns. All of these findings support

Table 1

Out-of-sample performance of the MVP constructed using covariance matrices estimated by conventional PCA, ICcPCA_I, ICcPCA_F, and POET.

Method	STD	IR	SR
ICcPCA_I	0.008	0.120	0.112
ICcPCA_F	0.008	0.151	0.143
PCA	0.010	0.107	0.099
POET	0.009	0.148	0.140

that ICcPCA is appropriate for analyzing stock return data with latent group structures.

4.2. Constructing minimum variance portfolio

Constructing the global minimum variance portfolio (MVP) is another important application of ICcPCA due to its ability to correctly identify both the homogeneity and sub-homogeneity when estimating covariance matrices of stock returns. The application of constructing an MVP has been well studied and discussed (see, e.g., Engle et al., 2019; Chen et al., 2019; Wang et al., 2021), and various estimation approaches for the covariance matrices of stock returns are developed. Our interest in this section is the numerical results of the out-of-sample performance of the global MVP constructed using the covariance matrices estimated by ICcPCA. Recall that stock returns of 160 companies from 8 industries are collected for 252 trading days. We use rolling windows with a size of 110 days (5 calendar months) to obtain the out-of-sample performance of the portfolio.

Similar to Wang et al. (2021), we compute the standard deviation (STD), the information ratio (IR), which is the ratio of mean return to STD, and the Sharpe ratio (SR), which is the ratio of mean excess return to STD. The numerical results in Table 1 compare the portfolios constructed with different estimates of the covariance matrices. As depicted, both ICcPCA_I and ICcPCA_F can achieve slightly better estimates of the covariance matrix, since the portfolios corresponding to ICcPCA_I and ICcPCA_F have the smallest STD. In addition, the global MVP constructed using ICcPCA_F has the highest IR and SR. As a result, ICcPCA works well when constructing the global MVP since the sub-homogeneity representing industrial effects can be well estimated.

5. Discussion

To conclude, we introduce a novel ICcPCA with a LOO-PCR clustering approach to study the market-wide effect (homogeneity) and latent industry structures (sub-homogeneity) in high-dimensional stock return data. Our numerical simulations confirm that the conventional PCA can only extract the homogeneity from the data, whereas ICcPCA not only provides a more accurate estimate of the common components, but also identifies the group-specific features, even for the group with small variations. Our empirical study demonstrates that, when applying ICcPCA, we can capture the industry information after removing the market effect. Additionally, the applications of ICcPCA are not limited to revealing the industry structure from the stock return data. ICcPCA can also be applied to any other high-dimensional data sets where a group structure exists but hides in the homogeneity.

Acknowledgment

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Appendix A

Algorithm 1 (ICcPCA).

1. Initial Step:

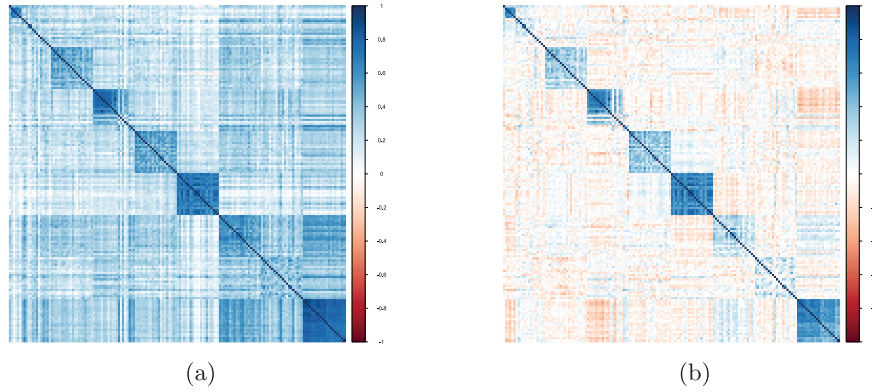


Fig. 4. Correlation plot for (a) the original stock data X , (b) the complement of the stock data $X^c = X - G_f \Phi_f^T$.

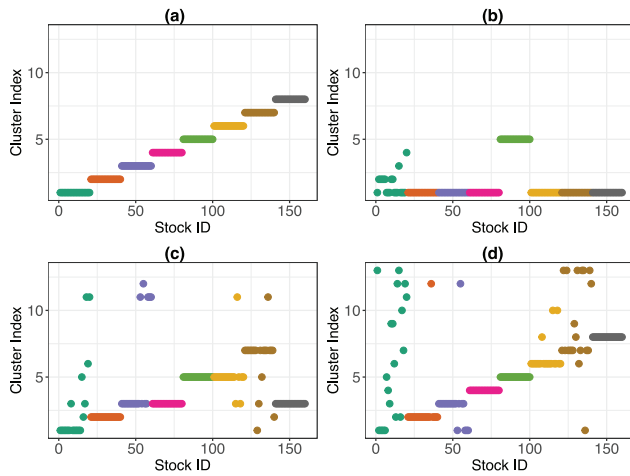


Fig. 5. Cluster membership for 160 stocks based on (a) their industry, (b) hierarchical clustering of the original data, (c) initial clusters obtained from ICcPCA, (d) final clusters obtained from ICcPCA. Stocks with the same color are from the same industry.

- (a) Perform PCA directly on the entire data X and select the number of components using a ratio-based estimator.¹ The resulting eigenvectors Φ_0 and PCs G_0 are served as the initial estimates of Φ and G . Then, find the initial complement $X_0^c = X - G_0 \Phi_0^T$.
- (b) Perform hierarchical clustering² for X_0^c based on a similarity matrix given by the absolute value of the empirical correlation matrix of X_0^c . The obtained clusters $C_0^{(j)}, j = 1, \dots, J_0$ are served as the initial clusters.

2. Iterative Step: for $s = 1, 2, \dots$

- (a) Cluster the variables into J_{s-1} groups according to $C_{s-1}^{(j)}$ and define variables from j th cluster as $X_s^{(j)}$. Perform PCA

on each cluster of variables $X_s^{(j)}$ and denote the obtained PCs and eigenvectors as $\Psi_s^{(j)}$ and $\Pi_s^{(j)}$, respectively for $j = 1, \dots, J_{s-1}$. Combine these PCs as $\Psi_s = [\Psi_s^{(1)}, \dots, \Psi_s^{(J_{s-1})}]$ and form a $\sum_{j=1}^{J_{s-1}} p_j \times \sum_{j=1}^{J_{s-1}} r_j$ block-diagonal matrix of eigenvectors as $\Pi_s = \text{diag}(\Pi_s^{(1)}, \dots, \Pi_s^{(J_{s-1})})$. Perform a further step of PCA on Ψ_s , and define the PCs as G_s and eigenvectors as H_s . Then, write $\Phi_s = \Pi_s H_s$ and compute the updated complement $X_s^c = X - G_s \Phi_s^T$. Due to space limits, details of the estimation procedure are left to the online supplementary material.

- (b) Perform leave-one-out principal component regression clustering (LOO-PCR clustering) for variables in X_s^c :

- (i) For each $k = 1, \dots, p$, leave the k th variable out of X_s^c .
- (ii) Group the rest of the variables of X_s^c based on $C_{s-1}^{(j)}$, perform PCA on each cluster again and denote the obtained components as $F_{s-1}^{(j)}, j = 1, \dots, J_{s-1}$.
- (iii) Fit J_{s-1} PCRs by using k th variable of X_s^c as the response and $F_{s-1}^{(j)}$ as the predictor for each $j = 1, \dots, J_{s-1}$, respectively.
- (iv) Compute the sum of squared residuals (SSR) for each J_{s-1} PCR model. Assign the k th variable to the cluster with the minimum SSR.³ Update the cluster index for k th variable in $C_{s-1}^{(j)}$.
- (v) Repeat (i)~(iv) for each k and denote the updated clusters as $C_s^{(j)}$.

- (c) Repeat (a) and (b) within this step until the clusters converge, and define this final converged cluster as $C_f^{(j)}$.

3. Final Step:

- (a) Repeat (a) in the Iterative Step, but using cluster $C_f^{(j)}$. The final complement is denoted by $X_f^c = X - G_f \Phi_f^T$.

¹ The ratio-based estimator (Lam and Yao, 2012) of the number of principal components r_c is defined as $\hat{r}_c = \text{argmin}_{1 \leq i \leq R} \frac{\hat{\lambda}_i}{\hat{\lambda}_{i+1}}$, where $\hat{\lambda}_1 \geq \dots \geq \hat{\lambda}_p$ are eigenvalues of the sample covariance matrix of data X and $r \leq R \leq p$ is a constant. In addition, an iterative estimation procedure proposed by Peng et al. (2025), which is a generalization of the original ratio-based estimator, is also considered to estimate the sub-homogeneity within each cluster correctly

² Average-linkage is used to measure the dissimilarity between clusters and the number of clusters are determined by the largest increment in height h_j in an agglomerative way (Bühlmann et al., 2013), such that $J_0 = \text{argmax}_{j \leq J_m} (h_{j+1} - h_j)$, where J_m is the maximum number of clusters to be considered

³ We assign the k th variable to the cluster that achieves the minimum SSR or equivalently the maximum $r^2 = SSR/TSS$, where TSS is the Total Sum of Squares. However, when the maximum r^2 is smaller than a threshold η , we treat the k th variable itself as a cluster, because it cannot be well-predicted by any clusters of variables. To prevent overfitting, η should not be large. In simulation studies and real data applications, we set $\eta = 0.05$ and the estimated numbers of clusters are close to their population counterparts. Alternatively, a penalty term τ on the number of clusters may also be considered to prevent the number of clusters from being too large, i.e. the k th variable $X_s^c(k)$ is assigned to cluster j , $j = \text{argmin}_j [SSR(X_s^c(k), F_{s-1}^{(j)}(k)) + \tau \log(J_{s-1} + 1)]$, $j = 1, 2, \dots, J_{s-1} + 1$, where τ can be determined by a cross-validation approach minimizing the mean squared recovering error

- (b) Cluster variables of $\mathbf{X}_f^c$ based on $C_f^{(j)}$, perform PCA on each cluster $\mathbf{X}_f^{c(j)}$ and denote the obtained cluster-specific components as $\mathbf{F}_f^{(j)}, j = 1, \dots, J_f$.

Appendix B. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.econlet.2025.112611>.

Data availability

Data will be made available on request.

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